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Summary

Artificial Intelligence graduate from NUML, isb with hands-on experience in developing end-to-end ML pipelines and building generative AI applications using RAG with a focus on reproducible workflows and scalable system design. Winner of Vyrothon 2026 AI Hackathon, where I contributed to building an AI voice agent for real-time lead automation. Skilled in FastAPI, Docker, and MLflow, with a focus on scalable and practical AI solutions. Keen to contribute to real-world projects in a collaborative environment.

Achievements

• Winner – Vyrothon 2026 (AI Hackathon)

- Secured 1st place among 500+ applicants through a competitive 3-stage selection process involving individual evaluation and team-based final rounds.
- Designed and developed an AI voice agent for real estate lead automation, enabling instant lead engagement, qualification, and appointment booking without human intervention.
- Built the system under strict time constraints, demonstrating rapid prototyping, problem-solving, and collaborative development in a high-pressure environment.
- Invited to speak at GDGOC,COMSATS webinar to share insights and strategies for winning hackathons.

Work Experience

• AI/ML Intern | SiberKoza Alpha (NASTP)

July 2025 – Sep 2025

- Developed and deployed end-to-end ML pipelines processing 80K+ records, covering data ingestion, feature engineering, model training, and evaluation, achieving 88% prediction accuracy while applying MLOps best practices with Docker, MLflow, and DVC for reproducible workflows.
- Built and served production-ready predictive models via FastAPI, tracking 40+ experiments with MLflow and reducing model deployment time by 60%, enabling automated versioning and real-time monitoring across 3 project verticals.
- Applied strong problem-solving and system design skills in high-pressure environments, translating complex requirements into scalable ML solutions.

Education

BSc Artificial Intelligence | NUML, Islamabad

2022 – 2026

- Relevant Coursework: Machine Learning, Deep Learning, Artificial Neural Networks, NLP, Computer Vision

Projects

• Final Year Project – AI-Based Virtual Try-On System

[\[GitHub\]](#)

- Developed an AI-powered virtual try-on system using diffusion models and computer vision for end-to-end garment fitting.
- Implemented preprocessing pipeline (body segmentation, pose estimation) with diffusion-based garment warping for realistic results.

• PurchasePulse – Real-Time Purchase Intent Prediction

[\[GitHub\]](#)

- Built a production-style ML system for e-commerce purchase intent prediction, serving via FastAPI with batch and single-inference endpoints.
- Trained XGBoost on 66 features achieving 90.23% accuracy / 0.89 ROC-AUC; integrated MLflow tracking and Streamlit drift-detection dashboard.
- Containerized full system using Docker Compose with DVC-managed data/model versioning for reproducible workflows.

• Telco ChurnGuard – End-to-End MLOps Pipeline

[\[GitHub\]](#)

- Built enterprise-grade churn prediction pipeline on 7K+ customer records achieving 85% test accuracy with XGBoost, hyperparameter tuning, and Great Expectations data validation.
- Deployed as a Dockerized FastAPI app handling 200+ batch predictions daily, with CI/CD via GitHub Actions automating linting, testing, and image builds.

• Rag Insight Engine – RAG System

[\[GitHub\]](#)

- Built modular RAG system for document-based QA over PDFs and text using LangChain and ChromaDB.
- Implemented recursive chunking, embeddings, and retrieval (similarity vs MMR) for semantic search.
- Integrated LLM-based responses with strict context grounding and citation support to reduce hallucinations.
- Designed evaluation pipeline to test retrieval accuracy and answer quality across queries.

Technical Skills

- **Languages & Tools:** Python, SQL, Git, GitHub Actions, Docker, FastAPI, MLflow, DVC, LangChain, Lang graph, AWS (EC2, S3)
- **ML & AI:** Deep Learning, NLP, Generative AI, LLM's, XGBoost, scikit-learn, TensorFlow, PyTorch
- **Data & Databases:** MongoDB, PostgreSQL, pandas, NumPy, ETL Pipelines

Certifications

- Supervised Machine Learning: Regression and Classification
- Oracle Cloud Infrastructure AI Foundations Associate
- Oracle Cloud Infrastructure Generative AI Professional

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